

CS543(Web Services) - Assignment 3

Integrate an External SOAP Partner

Team Id: 30

Team Members:

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1. Context

CampusEats integrates with an external payment partner, CampusPay Gateway, to process payments for customer orders through the charge operation. The payment edge uses SOAP because payment processing benefits from a strong, formally defined contract and structured message exchange between CampusEats and the external gateway. SOAP also provides support for message-level security and reliable, transaction-oriented communication, which are important for handling payment operations. The rest of CampusEats uses REST for simpler application-level communication, while the payment partner is isolated behind the SOAP integration boundary.

2. HTTP Binding

The CampusEats payment request is sent to the CampusPay Gateway using an HTTP POST request. The endpoint and SOAPAction are taken from the PaymentBinding defined in partner.wsdl.

Endpoint: <https://api.campuspay.example.com/soap>

Method: POST

Host: api.campuspay.example.com

Content-Type: text/xml; charset=utf-8

SOAPAction: ["http://example.com/campuspay/charge"](http://example.com/campuspay/charge)

HTTP-request

POST /soap HTTP/1.1

Host: api.campuspay.example.com

Content-Type: text/xml; charset=utf-8

SOAPAction: "<http://example.com/campuspay/charge>"

The SOAP envelope from soap-request.xml is sent as the body of this HTTP request. The request contains the payment credentials in the SOAP Header and the charge operation with the required payment details in the SOAP Body.

This is probably the better style for your assignment: **clear, simple, and enough to demonstrate that you understand the HTTP binding.**

3. Discovery

CampusEats would discover the CampusPay Gateway through a partner catalogue or registry maintained by the organization. The registry would contain the business name, service name, SOAP endpoint, and a WSDL reference describing the service contract. When CampusEats needs to integrate with the payment gateway, it can look up the partner in the registry and retrieve the WSDL from the registered location. A live UDDI server is not required; the catalogue entry serves as the registry record for this design.

Catalogue / Registry Entry

Field	Value
Business	CampusPay Gateway
Service	CampusPay Payment Service

Endpoint `https://api.campuspay.example.com/soap`

WSDL/tModel-style pointer `https://api.campuspay.example.com/soap?wsdl`

4. Fault Mapping

When the CampusPay Gateway rejects a payment, it may return a SOAP fault with the partner-specific error code card_declined. CampusEats handles this fault in its integration layer and maps it to the error defined by its own contract, payment declined. This prevents the CampusPay-specific error vocabulary from being exposed to CampusEats consumers.

Flow:

CampusPay Gateway



card_declined



CampusEats Integration Layer



payment declined



CampusEats Consumer

